

John B - This is RUSH!

2000 Thousand Oaks Dr.
Burleson, TX 76029
October 17, 1977

Send copy to John Craig
But start typesetting, etc
immediately -

- W -

Mr. Wayne Green, Publisher
KILOBAUD Magazine
Peterborough, NH 03458

\$150

Dear Wayne:

Here, for your consideration, is my review of the Radio Shack
TPS-80. Assuming anyone who has enough curiosity about personal
computing to have purchased a computer magazine already has some
familiarity with basic terms such as ROM and RAM, I've tried to
write it: 1. Based on that assumption. and 2. Without going into
so much explanation that I was doing the job the instruction
manual should do if and when the reader buys a TPS-80.

I'm including photos also, but make no mention of them in the
text, so you can include them as desired, with appropriate
headings. One, I think, has the CLOAD asterisks in the upper
left corner of the screen, which might bear noting.

I wrote Radio Shack, special delivery, last week, told them what
I was doing, and requested any photos or information they could
supply...or at the least their permission to break the seal, open
the box and photograph the board without voiding the warranty.
They didn't see fit to reply, so I voided the warranty anyway.
Keep your fingers crossed for me that it doesn't quit working soon.!

Hope you like it...I'll be interested in your thoughts and/or
suggestions.

Sincerely,

Ed Juge

Ed Juge

Ed Juge
2000 Thousand Oaks Dr
Burleson TX 76028

525
~~AN OWNER'S EYE-VIEW OF THE RADIO SHACK TRS-80~~

By Ed Juge

Several years ago, my business purchased a Hewlett-Packard 9830. H-P called it a programmable calculator, but it was a micro-computer with H-P BASIC in ROM, and it was slightly larger than an electric typewriter. We plugged it in, turned it on, and it was an up-and-running computer. Price, just about \$14,000. I knew when I bought an Altair 8800 a couple of years ago that it wouldn't be quite that easy, ~~that~~ after I built and debugged it, there would still be a lot of accessories to buy keyboard, output device, etc. and after they were all operating I'd still have to make them work together, teach ~~it~~ BASIC, and spend some time working with it before I could actually run in BASIC. Not having a background in computer hardware or digital electronics, and finding people who did few and far between, ~~though~~ it took more time than expected. Through the efforts of several friends, and one in particular, the 8800 finally ran in BASIC about a month ago, ~~and~~ some \$1,500 after my initial decision to buy. All of this has nothing to do with the Radio Shack TRS-80, except to explain why I was so excited when I heard of its introduction, and why I ordered immediately and received one of the first 60 or so off the production line. You see, kits and assemble-your-own-system computers are beautiful for hardware buffs, but my one and only interest is programming.

The first surprise from Radio Shack was delivery within six days of the original ⁽¹¹⁾expected delivery⁽¹¹⁾ date. Astounding for a new item, whose target date was set before a production line even existed.

The TRS-80 came well packed, and consisted of the keyboard, video monitor, cassette recorder, AC power pack for the computer, and two tapes, one blank and the other containing the prerecorded games Blackjack and Backgammon. Also included was a 30-page preliminary instruction book, and a card saying that the 300-page owner's manual would be mailed to me in a few weeks.

Second surprise...everything plugged together easily; I connected power, threw the switches, typed in NEW, and it was a complete up and running system, in BASIC...*the first time!* Just like my \$14,000 H-P had been, but for ~~the amazing price of only~~ \$599.95!

The TRS-80's CPU, 4K ROM, 4K RAM, Video and cassette interfaces, everything, is housed in the keyboard enclosure. No plug-in boards. No mother board. No heat problem. No twenty-pound power supply. The TRS-80's external power supply is hardly larger than a large capacitor we had to add to the Altair to make it handle an extra memory board. The housing is beautifully styled in gray and black. The video monitor is in a matching cabinet, making an unusually attractive setup which wouldn't be out of place in any setting. There is a small door in the rear of the keyboard enclosure which covers a 40-pin connector for later addition of peripheral equipment, should you wish to expand your TRS-80 system. A RESET button is also under this door. It permits you to "unlock" any condition (such as an unending loop) which can't be unlocked from the keyboard BREAK key. RESET unlocks without disturbing your memory.

The ~~keyboard~~ keys look like they were taken ~~off~~ of an IBM Selectric typewriter, and the touch is excellent. Either a couple of them aren't as positive as they should be, or I get a bit light-fingered on them, ~~though~~, as I occasionally miss them when typing fast.

Radio Shack was thoughtful enough to provide a ~~really~~ nice cassette recorder with its own built-in ~~AC~~ power supply...no batteries to buy unless you want to. The TRS-80 starts and stops the recorder ~~when~~ you give it, either, CLOAD or CSAVE. Should you want to rewind or run fast forward, you must unplug the "remote" cable to the TRS-80. An override momentary-contact button on the recorder would have been more convenient, and saved wear and tear on the tiny jacks and plugs.

The video monitor is a 12-inch unit with very good display, and is easy to read under any room-lighting conditions. If you're ~~saying~~ that a \$199.95 video monitor is quite expensive just for a matching cabinet, ~~though~~...well, maybe so, until you rush home from work anxious to do some computing, only to find the portable TV you've modified, being used by a room full of kids to watch the GONG SHOW. You won't have to worry with the TRS-80's monitor, it has no tuner, so won't tune in TV. Your output device will stay an output device.

OPERATION

Up until now, I have never seen a hobby computer you could carry home from the store, plug in and use in anything other than machine language (numeric codes). The manufacturers apparently have been so deeply into computers themselves, that they failed to realize the

~~number~~ of people out there in the real world ~~who~~ have no experience or training with computers, and ~~who~~ aren't interested in constructing a unit, or ~~in~~ having to program with lights and switches, or even numeric codes...but who would jump at the chance to own a computer they could talk to in near "plain English" languages such as BASIC. There certainly is a market composed of hardware buffs with training and experience in the field, but how many more of us must there be ~~composing~~ the non-technical part of the market? Well, obviously, Radio Shack is going to find out, because we're the ones they've built the TRS-80 for...along with small businesses and the educational market.

In shooting for our market, however, there is one essential ingredient other than an up-and-running machine... training the complete novice to program a computer.

Radio Shack's preliminary manual is one giant step in the right direction. If their 300-page book is an expansion of the small one, they should have the problem solved. The preliminary manual not only gives you complete initial hookup instructions, but it gives you a ~~rather~~ thorough course in the basics of programming. They start with simple arithmetic calculations, and progress through each of the TRS-80's functions, commands and statements. Every step of the way, the manual spells everything out in simple, non-technical language, and it gives you sample programs to try as you read, so you will see for yourself how the TRS-80 operates, given a particular command. My fifteen-year-old went through the entire thing with no trouble or

questions arising that he couldn't answer himself. ~~Radio Shack~~ indicates ~~in~~ the preliminary manual that the 300-page book will contain a detailed, complete course ~~in~~ programming in *Radio Shack Level I BASIC*.

If you're interested in seeing the TRS-80 run immediately, without having to go through this short course in programming, their game tape will have you playing Blackjack or Backgammon in a couple of minutes.

The TRS-80, in operation, appears to be highly reliable and trouble-free. Mine has been on for a period of 17 hours straight at least once, and I've probably logged a total of 40 hours minimum so far. During that time, operation has been flawless. To offer a comparison, I've seen the time when BASIC had to be loaded into the Altair (via paper tape) ten times before it was done without some type of error showing up in the operation. The H-P 9830 could be expected to show some kind of problem in that many hours (H-P's salesman once ~~told~~ us that they anticipated one error in something like every 1.5 million calculations, I ~~think~~ it was. These weren't errors in calculations, but in tape loading or storing, memory dumps, etc.). So, the TRS-80's operation ~~thus far~~ has been ~~an~~ outstanding ~~one~~ by these standards. I've found no problems from power-line surges (as when the air-conditioner comes on), dust, carpet fuzz, or anything of this nature, to which some units are super-sensitive.

Radio Shack might look into some additional ~~RF~~ filtering, however, as my TRS-80 does slightly affect the color on a TV, in the next room, which is operating on rabbit ears. I suspect (because the level of interference is so low) that a good outdoor TV antenna would cure ~~it~~.

The advertised 4K of RAM, my computer tells me, is exactly 3,583 bytes. This is more than enough for all but the most involved and lengthy programs. If printed out on a typewriter, a 4K program would be ~~around three feet in length, to give you an idea.~~ Should you anticipate needing more memory, additional plug-in memory is available for your TRS-80, at your local store.

Most hobby computers do nothing when first turned on, except operate in binary codes. Radio Shack's 4K BASIC in ROM makes the TRS-80 an intelligent and conversational machine ~~immediately when~~ power is applied. If you've experienced the other ~~kind~~ of operation, you're in for a pleasant surprise!

RADIO SHACK'S LEVEL I BASIC

Radio Shack's 4K "Level I" BASIC appears to pack an awful lot into 4K. It includes most standard basic commands, video handling, and graphics. It does not, however, include trig. functions. Commands available are NEW, LIST, RUN, CONTinue, REMark, LET(optional), GOSUB - RETURN, FOR-NEXT-STEP, GOTO, IF-THEN, INPUT, ON...GOTO, ON...GOSUB, PRINT, CSAVE, CLOAD, DATA, READ, RESTORE, STOP, and END. Functions are: ABS, RND, INT, TAB, +, -, X, $\div$, >, <, <=, >=, and =. The command MEM ~~is also~~ available which tells you the amount of RAM which remains unused at any

time,

Graphics are available, and their special commands are: CLS (clear screen), SET(X,Y)(turns on point X,Y), RESET(X,Y)(turns off point X,Y), and POINT(X,Y) (used with IF...THEN, and determines if X,Y is set).

The non-graphics display is made up of 16 ~~lines~~ of 64 characters ~~per~~ line, and this display is mapped with 1,024 positions. You can instruct the TRS-80 to PRINT AT (position #), to place your display anywhere on the screen you wish without the need for printing a bunch of blank spaces or lines to get there. The graphics display consists of 128 horizontal and 48 vertical positions, and of course graphics and text can be used together.

The TRS-80 has only three error codes:

WHAT — meaning the computer does not understand your instruction.

HOW — meaning the instruction is understood but cannot be executed
(such GOTO a nonexistent line number.)

SORRY — meaning you've used all of the available memory.

When you've programmed in an error, and you RUN the program, the computer will reach the error, stop, and display the error code followed by the line in which the error occurs. A question mark appears in the faulty line at the exact position of the error, making debugging a program extremely simple and straightforward.

Only two string variables, A\$ and B\$ are available in Level I, and each is limited to 16 characters. The single array A(n) can be used,

and is limited only by the memory capability of your TRS-80. Variables A through Z are permitted. Integer limit is 32,767, which is also the limit of line numbers. Large or very small numbers are handled in scientific notation.

Most program statements and commands can be abbreviated, and statements can be stacked on a single line. Thus a program which reads:

```
10 FOR X = 1 TO 100
20 PRINT X, 2X, X/2
30 NEXT X
40 END
```

could be written as:

```
10 F.X=1TO100:P.X,2X,X/2:N.X:END
```

(END has no abbreviation). The "long" version requires 51 bytes of memory, where the abbreviated version takes only 33 bytes...a saving of 18 bytes or 35%. This can be significant in lengthy programs.

One of the most interesting features of the TRS-80 is its excellent random number generator. A random number is generated by the command RND(X), with a number being returned between 0 and X. A random number between 0 and 1 will be produced by the command RND(0). The numbers returned are truly random, and no repeating sequences have been observed, even from initial RUN of a RND command. Combining RND commands with the graphics capability of the TRS-80 can provide hours of fun. A few samples of such programs are included in the preliminary manual.

A cursor "_" is always present on the monitor, indicating where the next display will appear. During programming, it can be backspaced with a "←" key to correct errors within that line. Correcting a line after entry requires retyping the entire line. Eliminating a line is done by typing the line number only and ENTER. The TRS-80 puts all lines in proper sequence, regardless of the ~~sequence~~ in which they are entered.

The LIST command, puts the program (first 16 lines ~~of it~~) on the screen, and a "↑" key scrolls it up for viewing additional lines.

There are two additional keys, related to the two mentioned above. These are "→" and "↓", both of which are inoperative. The preliminary manual does not mention them, although it would have saved Radio Shack a lot of phone calls from customers who are afraid they have a defective unit. My call to them confirmed my suspicions that these keys are for use in their 12K BASIC, which should be available before the end of 1977. This was the only thing ~~which~~ I felt should have been included in the preliminary manual but wasn't. The 12K BASIC will be available in plug-in internal ROM, and is expected to cost around \$100, installed by your local Radio Shack.

A complete listing of Level I Basic is contained in the back of the manual, along with an explanation of each command, and its abbreviation.

For those of you who worry about your computer "going to sleep", Radio Shack has included in the upper corner of the display, a couple

of asterisks ~~which~~ appear during CLOAD. If the cassette is loading properly, one of them will blink on and off. When you're storing a program on tape (CSAVE), recording volume level is automatic. When loading a program from tape into the TRS-80 (CLOAD), the recorder's volume control operates. It's setting should be (according to the manual) between 7 and 8. I found this critical on their pre-recorded game tapes, as their level is somewhat low. On programs I've recorded on tape, playback is perfect at any volume setting above 3.

OPERATING IMPRESSIONS

Using any micro for a while is bound to leave one with predetermined ideas about what a good machine should and shouldn't do...features and commands it should ~~have~~, or ~~it~~ shouldn't. These ideas are based largely on an individual's use of a computer. Trying to allow for my preconceived ideas (and hoping 12K basic will include some of them), I ~~have to~~ say that ~~without even~~ considering the low price, the TRS-80 has to run at the top of the list for anyone wanting an up-and-running, no-hassle, nothing-to-build, nothing-else-to-buy personal computer. It runs in BASIC as it comes from the carton. Reliability couldn't be better. Size makes it useful anywhere. Styling would be ~~most~~ difficult to improve on. Assuming Radio Shack plans to arrange for at least limited local (plug-in) service, finding a Radio Shack store ~~locally~~ should be easy in most places. Where else can you get an up-and-running machine with a manual which can teach even the complete newcomer, how to program?

There's always the question of speed in a computer...

To most of us, the only reason to speed up a machine capable of about 5,400 simple calculations per minute (that's around 90 per second) would be to boost our ego. I can think of few practical applications of a hobby computer which would be disadvantaged by that speed. That is approximately the speed of the TRS-80.

K/LOBAUD published the results of some benchmark timing tests in issue #10, comparing 33 microcomputers' speeds. In case you're interested, the TRS-80 stacks up as follows:

Test #	Seconds	Test #	Seconds
1	2.5	5	46.0
2	18.6	6*	67.5*
3	24.4	7	110.1
4	37.0		

* Test #6 included a DIM statement which is not used on the TRS-80.

A REM statement was substituted.

This places the TRS-80 in the slower third of the machines tested, while other computers using the same Z-80 CPU chip scored in the top third. The difference could be a combination of the clock speed Radio Shack uses in the TRS-80 and/or their Level I Basic. It's possible that their 12K basic will show faster times. It's reassuring to know, however, that you are smart enough to give your computer a problem that it has to think about for a few seconds sometimes! Maybe man still has the upper hand over machine.

Many people ask, "So you've got a computer...whaddya do with it?"

This could be an embarrassing question, ~~sometimes~~. Everyone's answer will be different. Unless you're an engineer or businessman who can use one to solve equations or keep inventory records, the question of practicality of use is questionable. Don't let this discourage you...how really, truly practical is your television set most of the time?

My enjoyment in using a computer comes from programming: Solving problems...whether or not I have any use for ~~seeing them solved~~. It's fantastic exercise for your brain, which I enjoy. Just writing a program ~~which~~ runs is only half the fun for me. After my program runs, I like to see if I can make it run faster by altering or improving the programming, or make it do the same thing with fewer steps. I doubt that many people will ever find themselves writing the best, fastest possible routine on the first try. In THE BASIC FORUM (KILOBAUD #9), on page 18 of the magazine, a problem is offered: To write a program, asking you for a number (n). The program then locates and prints all prime numbers between 1 and n. My first try required over 30 seconds to search from 1 to ~~an n of~~ 100. My first revision cut the time in half. The best I could do was about 11.4 seconds without the printout containing an error because of my programming. I don't want to admit the time spent on it, but it was fun and occupied several interesting evenings of my time. I sincerely hope that will become a monthly undertaking in their column. They solicit reader's programs, and will publish the best they get, so you can see how your programming skills are shaping up.

Being a ham radio operator, I have used my computer to calculate all possible combinations of frequencies (channels, if you prefer) in the 2-meter FM band which will combine to produce an unwanted signal on any given (other) channel. The program is simple and short, but since you can have a multiple of one channel combining with a different multiple of another, if you check all possible combinations up to the fifth multiple of each, that simple program may take half an hour to run, and involve 9,000,000+ calculations. Try that with your pocket calculator! The results could be useful under certain circumstances, but writing the program was interesting.

Practical applications? Well, if you must feel your machine is practical, and if you don't think making it do useful work will ruin its standing as a "hobby" computer, you can have it keep your family financial records, store phone numbers, ~~on~~ addresses, or both, receipes, remind you of birthdays and anniversaries, almost anything. Just use your imagination. Radio Shack may be a step ahead of you on some of these uses, they have cassette program packages available for small-business payroll, teaching the kids math, kitchen uses (menus, conversion tables, directory and message center) and personal finance. These vary from a single cassette to a 7-cassette portfolio, and prices range from \$4.95 to ~~only~~ \$19.95. More is planned.

Just because the TRS-80 is a one-circuit-board unit without plug-in accessory boards, don't think you're stuck with what you've bought. They plan to have, ~~yet~~ this year, a couple of accessory printers, a disc memory, provisions for a second cassette recorder, and the

12K BASIC already discussed. Right now you can get additional internal plug-in RAM options of an additional 4K or 12K. Planned expansion also includes Serial I/O, Parallel I/O, an expansion unit for additional PC boards, and a modem. They're considering digital cassettes, enhanced graphics, a music synthesizer, external device controller, color graphics, a 14-key numeric pad, and who ~~really~~ knows what else.

Unquestionably, Radio Shack's TRS-80 has given a giant boost to their image, and provided ~~at last~~ a personal computer for the masses: trained or untrained. ~~For~~ those among you who say such a packaged unit, ..ready to run, requiring no hardware knowledge, limiting the haywiring you can do, isn't in the true computer-hobby spirit; ~~To you~~, I have one thing to say. Wanna buy an up-and-running 12K Altair at a bargain?

[To the rest of you!...anybody for a TRS-80 users' group? ]

Captions (Juge)

The radio shack TRS-80 displaying part of a program listing. The computer's ac power supply can be seen behind the video monitor.

TRS-80 during CLOAD. Note two asterisks in upper left corner of screen. The right one blinks on and off during cassette loading if program is loading properly.

Keyboard ^{with} ~~and~~ its circuit board, and partial rear view of TRS-80's main board.

Bottom view of TRS-80 shows board containing CPU, 4K ROM, 4K RAM, video logic, and cassette and video interface hardware. Plugs for cassette recorder, video monitor and ac power pack are at upper left with power on/off switch. Reset switch is at upper right next to 40-pin connector for peripheral equipment.

5215

OCTOBER 25TH, 1977

ED JUGE
2000 THOUSAND OAKS BLVD
BURLESON, TEXAS 76028

DEAR ED,

THANKS FOR THE TRS-80 REVIEW...IT'S IN THE WORKS. THE
PROOFS WILL BE ALONG SHORTLY FOLLOWED BY A CHECK FOR \$150.

THANKS AGAIN.

REGARDS,

JOHN CRAIG

C: John Barry

